

Assignment 1 Instructions

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Problem 1: Benchmark Stationarity Detection

Write a function named `is_stationary(samples: list[float]) -> dict[str, Any]` that determines whether benchmark execution times remained steady throughout a test run rather than drifting over time. The input list must preserve the exact chronological order in which measurements were captured.

Requirements:

1. Input Validation:

- If the input list contains fewer than `MIN_SAMPLES_FOR_STATIONARITY` (12) samples, return an `unknown` record stating that there are too few samples to divide into thirds.
- Compute the overall median of the entire sample list. If this median is less than or equal to 0, return an `unknown` record stating that the median is not positive.

2. Splitting and Segment Analysis:

- Let $k = \lfloor n/3 \rfloor$, where n is the total number of samples.
- Extract the first third (the first k items) and calculate its median.
- Extract the last third (the last k items) and calculate its median.

3. Drift Calculation:

- Calculate the raw drift (Δ) as:

$$\Delta = \text{median}(\text{last third}) - \text{median}(\text{first third})$$

- Compute the relative drift by dividing the absolute value of Δ by the overall median of the run:

$$\text{relative_drift} = \frac{|\Delta|}{\text{overall_median}}$$

4. Direction Classification:

- Identify the direction of drift as a string:
 - "slower" if $\Delta > 0$
 - "faster" if $\Delta < 0$

- "flat" if $\Delta = 0$

5. Output Record:

- Return a **measured** record where the primary boolean result is **True** if the relative drift is less than or equal to **STATIONARITY_TOL** (0.10, or 10%), and **False** otherwise.
- Include the metadata fields:
 - **first_third_median_ms**: Median of the first third, rounded to 4 decimal places.
 - **last_third_median_ms**: Median of the last third, rounded to 4 decimal places.
 - **drift_ms**: Raw drift (Δ), rounded to 4 decimal places.
 - **drift_relative**: Relative drift, rounded to 4 decimal places.
 - **direction**: The classified direction string.
 - **tolerance**: The constant **STATIONARITY_TOL**.